



Research Knowledge Base and Reasoning

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Area (1/3)

- (i)* Language Standards and Extensions: it focuses on creating and standardizing the languages for knowledge representation.
- (ii)* Logic and Reasoning: it focuses on developing logic and reasoner.
- (iii)* Ontologies and Modeling: it focuses on building ontologies in several domains.
- (iv) Semantic Web Services: it focuses on using semantic technology in developing web services.



Area (2/3)

- (v)* Linked Data: it focuses on publishing the existing data to Linked Data.
- (vi) Matching and Integration: it focuses on proposing methods in integrating multiple resources and matching approaches to existing schema or vocabularies.
- (vii) Query Processing: it focuses on all topics in querying the Semantic Web data.
- (viii)* Security, Trust, and Provenance: it focuses on investigating the quality of the provenance and trustworthiness of sources.



Area (3/3)

- (ix) Knowledge Extraction and Discovery: it focuses on mining and finding a new knowledge of the Semantic Web data.
- (x) Search, Retrieval, and Ranking: it focuses on proposing methods in retrieving the Semantic Web data to obtain better results.
- (xi) User Interfaces and Annotation: it focuses on capturing user interaction for reaching goals of some tasks.
- (xii) Applications: it focuses on using the Semantic Web technologies for vertical applications in various domain use cases.



Source (1/2)

- Journal of Web Semantics:
<https://www.sciencedirect.com/journal/journal-of-web-semantics>
- Semantic Web Journal: <https://www.semantic-web-journal.net/issues>
- Applied Ontology:
<https://www.iospress.com/catalog/journals/applied-ontology>
- Knowledge-Based System:
<https://www.sciencedirect.com/journal/knowledge-based-systems>



Source (2/2)

- WWW: <https://dl.acm.org/conference/www>
- ISWC: <https://iswc2023.semanticweb.org/>
- ESWC: <https://eswc-conferences.org/>



A few examples

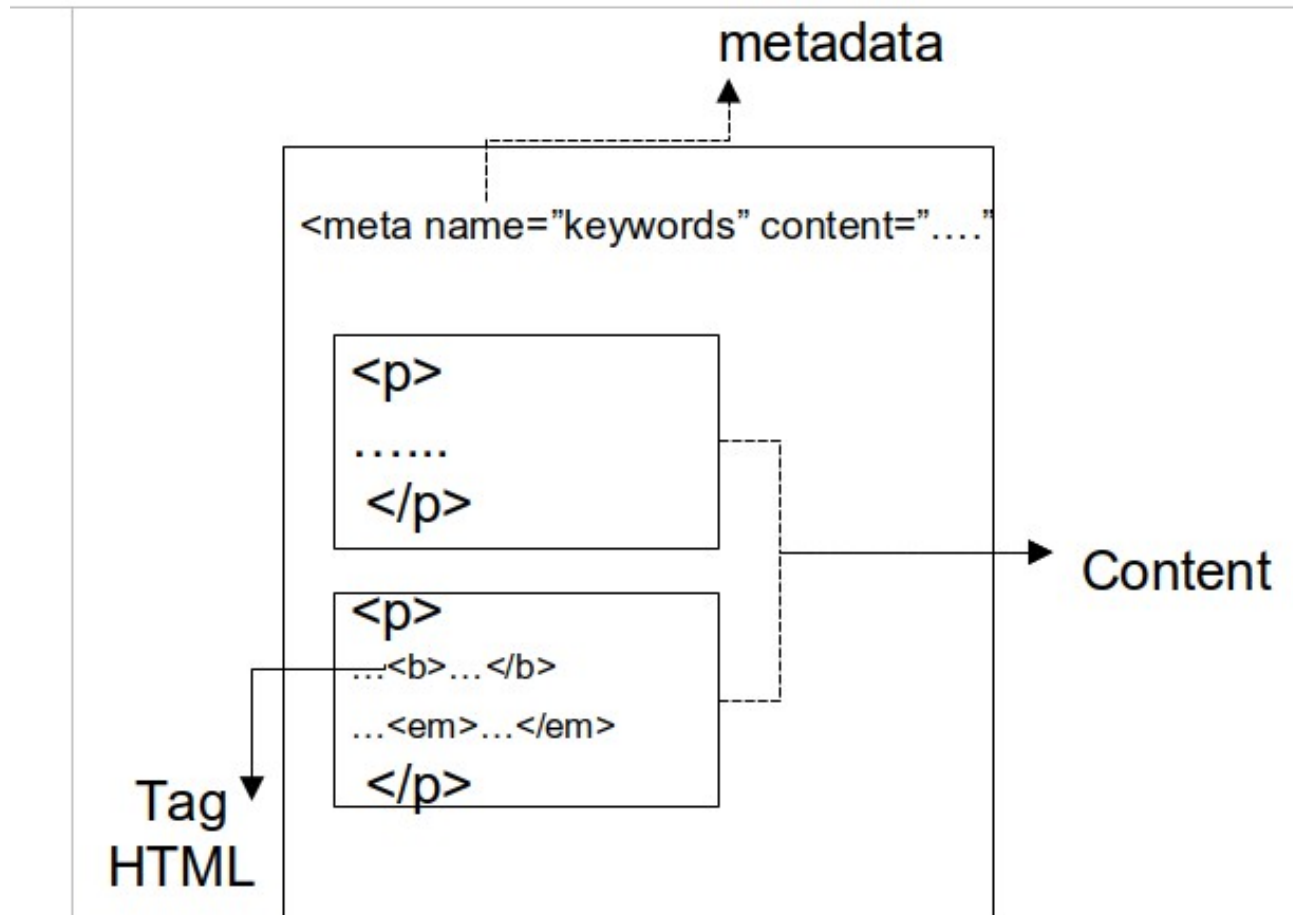
Dewi Wardani



Metadata

- “Pengembangan Automatic Text Summarization Berbasis Graf dengan Fitur Tag HTML dan Metadata”*
- Utilizing the concept of metadata to improve summarization (Language Processing)
- Adding the score of metadata and HTML tag into the existing method

*) Improving Graph-Based Summarization with HTML Tag and Metadata Features.





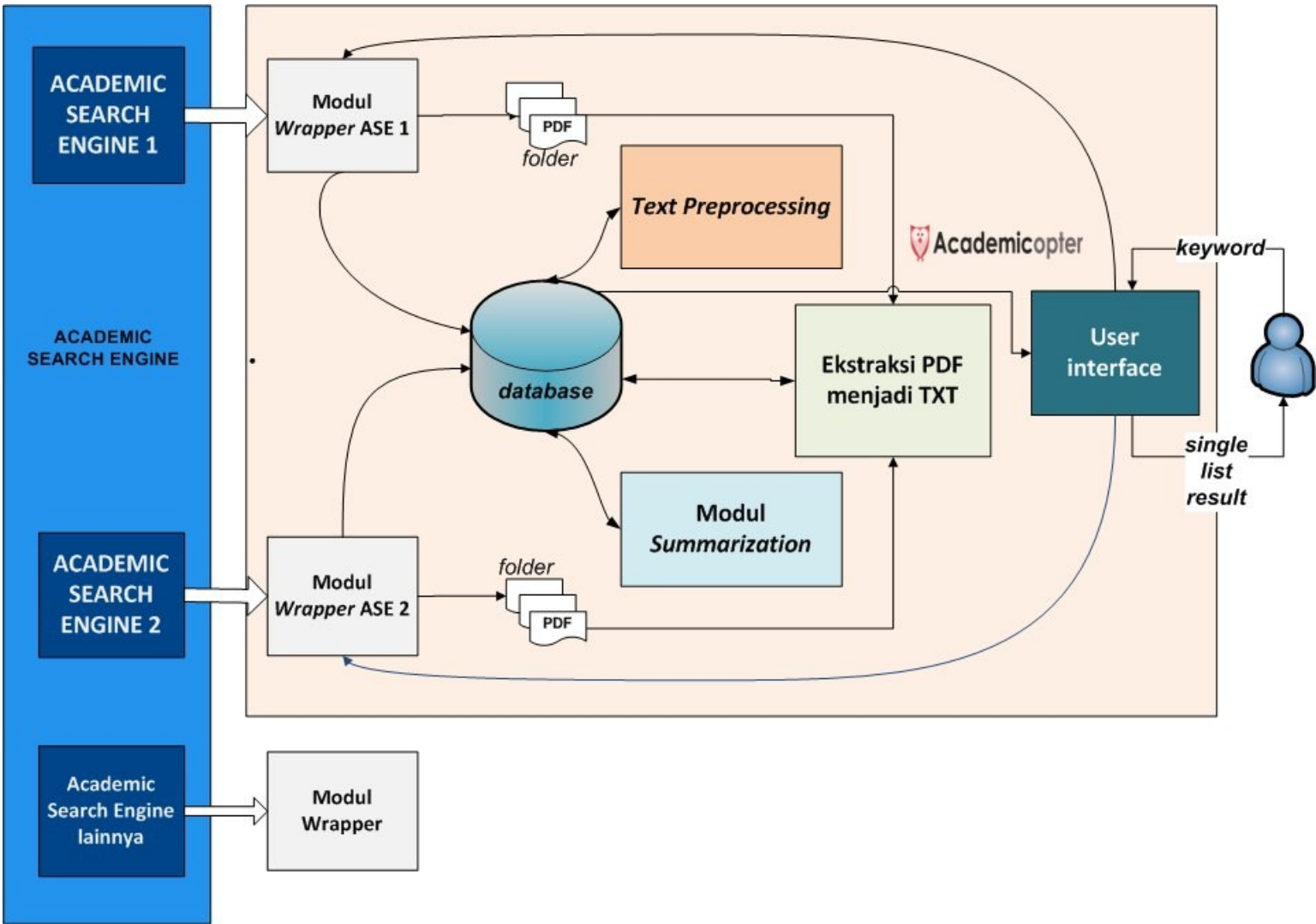
- Future work:
 - In case that webpages are semantic tagged with universal vocabulary such as schema.org
 - Knowledge base (ontology) might help to notice the focus of paragraph
 - Finding satire paragraph



Metadata

- “Academicopter : Mesin Pencarian Meta Untuk Akademik Dengan Peringkasan Otomatis Pdf Jurnal Ilmiah”*
- Utilizing the concept of metadata in meta search (Language Processing)
- Utilizing No.1 but its implemented in academic search engine

*) Academicopter: Metasearch with PDF Automatic Summary of Scientific Articles

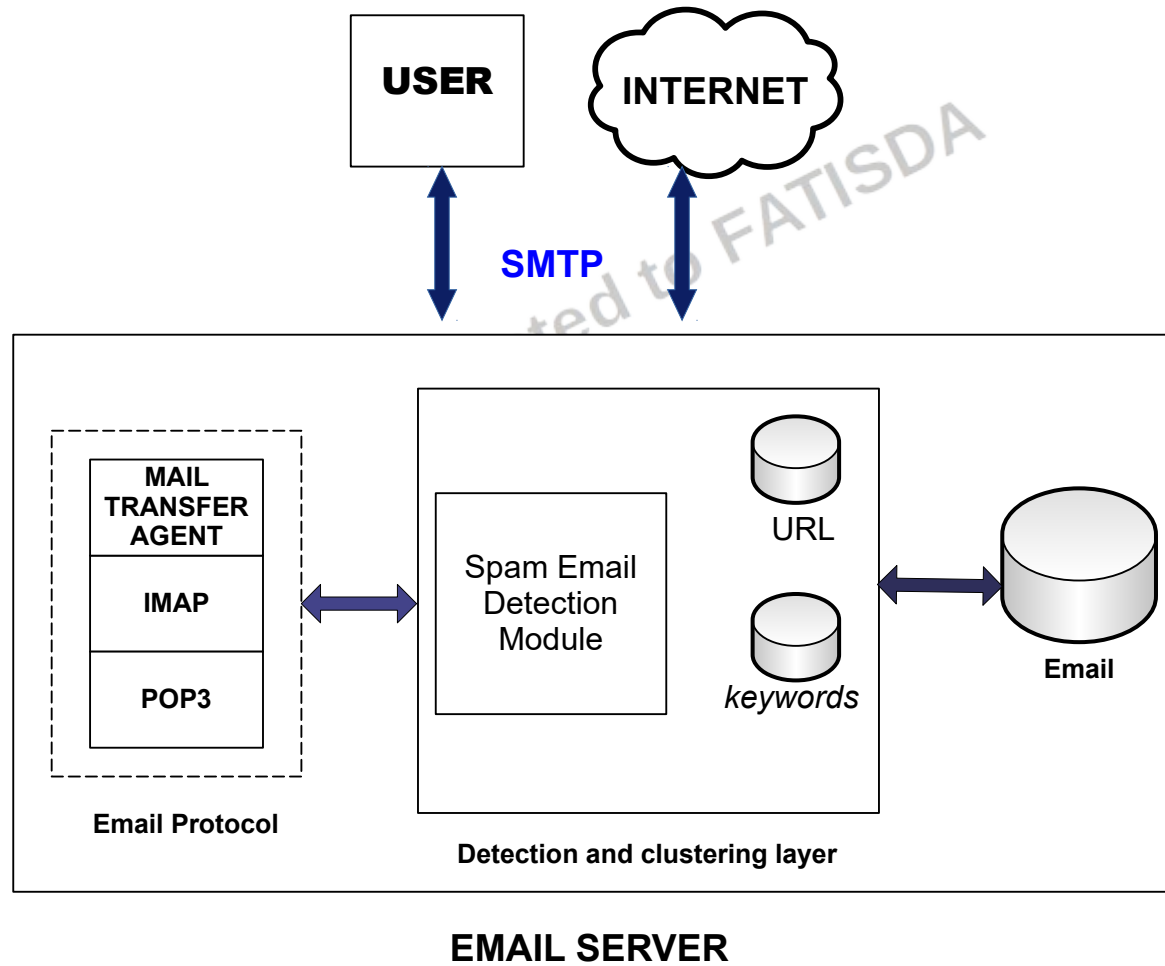




Metadata

- “Simulasi Deteksi Otomatis Spam Email Berkonten Pornografi”*
- Utilizing the concept of metadata to calculate the similarity of the other metadata (Language Processing)
- Proposed the framework to recognize spam emails by only comparing only the metadata

*) Using metadata in detection spam email with pornography content



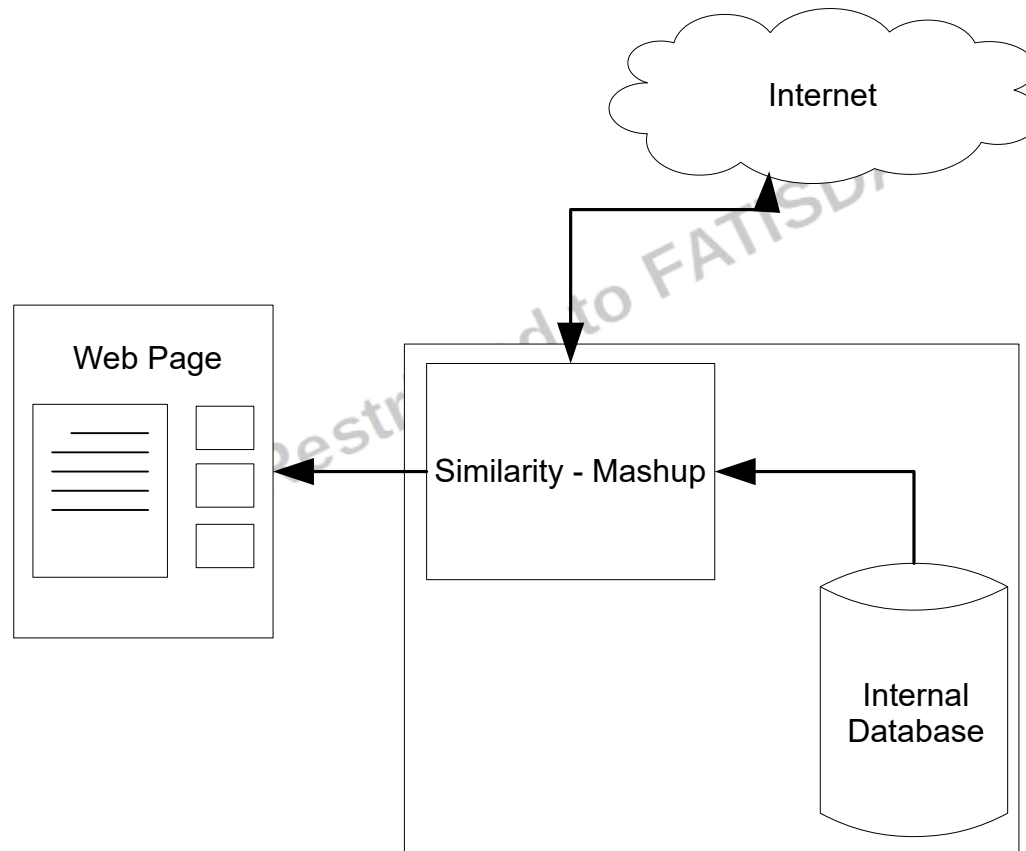


- Future work:
 - Improving the speed of the process of recognition
 - Study the connection among metadata
 - Addition in indexing technology
 - Broader domain of spam
 - Implementation in social media



- “Metode Similarity - Mashup Untuk Framework Modul Relevant Content Pada Content Management System (CMS)”*
- Utilizing the concept of metadata to calculate the similarity (Data Integration)
- Proposed the framework integrate data only based on the metadata. Alike special RSS. Add a new modul into existing framework

*) The Mashup Relevant Content Module for Content Management System (CMS)



- Future work:

- Integration data by using knowledge base (ontology) from several resources data type



- “Pendekatan Semantic Similarity Untuk Mendeteksi Kemiripan Ide Proposal Tugas Akhir Mahasiswa”*
- Adding summarization, finding entity over unstructured data. Using knowledge base in calculating semantic similarity
- Future Work
 - Better framework, by improving with graph approach

*) SSTI: Semantic Similarity to detect Novelty of Thesis Ideas



Data tidak terstruktur

Pasukan Sekutu memasuki kota Bandung pada bulan Oktober 1945. Mereka langsung menuntut pasukan Indonesia untuk menyerahkan senjata yang diperoleh dari Jepang. Mereka juga minta TKR mengosongkan kota Bandung bagian utara, paling lambat tanggal 29 Oktober 1945.

Rakyat Bandung tidak mematuhi ultimatum tersebut. Di bawah pimpinan Arudji Kartawinata, TKR melakukan serangan terhadap Sekutu. Akibatnya, terjadi pertempuran di Pasar Baru, Lengkong Besar, Cicadas, Andir, dan sekitar Gedung Sate.

RM Suwardi Suryaningrat lahir pada tanggal 2 Mei 1889 di Yogyakarta. Namanya lebih dikenal dengan Ki Hajar Dewantoro. Beliau mendirikan Taman Siswa tanggal 3 Juli 1922 dengan tujuan memajukan pendidikan bangsa Indonesia.

Data terstruktur

Date	Snippet
29 Oktober 1945	Mereka juga minta TKR mengosongkan kota Bandung bagian utara, paling lambat tanggal 29 Oktober 1945.
3 Juli 1922	Beliau mendirikan Taman Siswa tanggal 3 Juli 1922 dengan tujuan memajukan pendidikan bangsa Indonesia.



Ontology

- “Pemanfaatan Ontology Untuk Mendukung Hasil Association Rule Apriori”*
- Utilizing ontology to improve (representing data) the result of association rule Apriori
- Proposed the interestingness rule (IR) for hierarchical ontology

*) The Utilization of Ontology in Association Rule



- Future work:
 - Different style of ontology (more complex), proposing the new IR score measure

$$IR = \left[\log(A \times Trans(A)) + \log(B \times Trans(B)) \right] \times \left[\frac{Trans(A, B)}{Total_{trans}} \right]$$



- “Sistem Pendukung Keputusan Pemilihan Program Studi Dengan Pendekatan Tes Minat Bakat Menggunakan SWRL dan Metode SAW”*
- The using of Ontology and SWRL in data integration for DSS
- Future Work:
 - Improving the ontology by using inference and more intelligent agent

*) Major choosing decision support system with talent interest approach using SWRL and SAW method



- “FoFA: Penerapan Teknologi Semantik pada Aplikasi Berbasis Android Mengenai Informasi Diet untuk Penderita Autism”*
- Creating framework to use ontology directly in mobile application
- Future Work:
 - Using upper technology layer in mobile application.
 - Effective framework in using semantic tech in mobile application

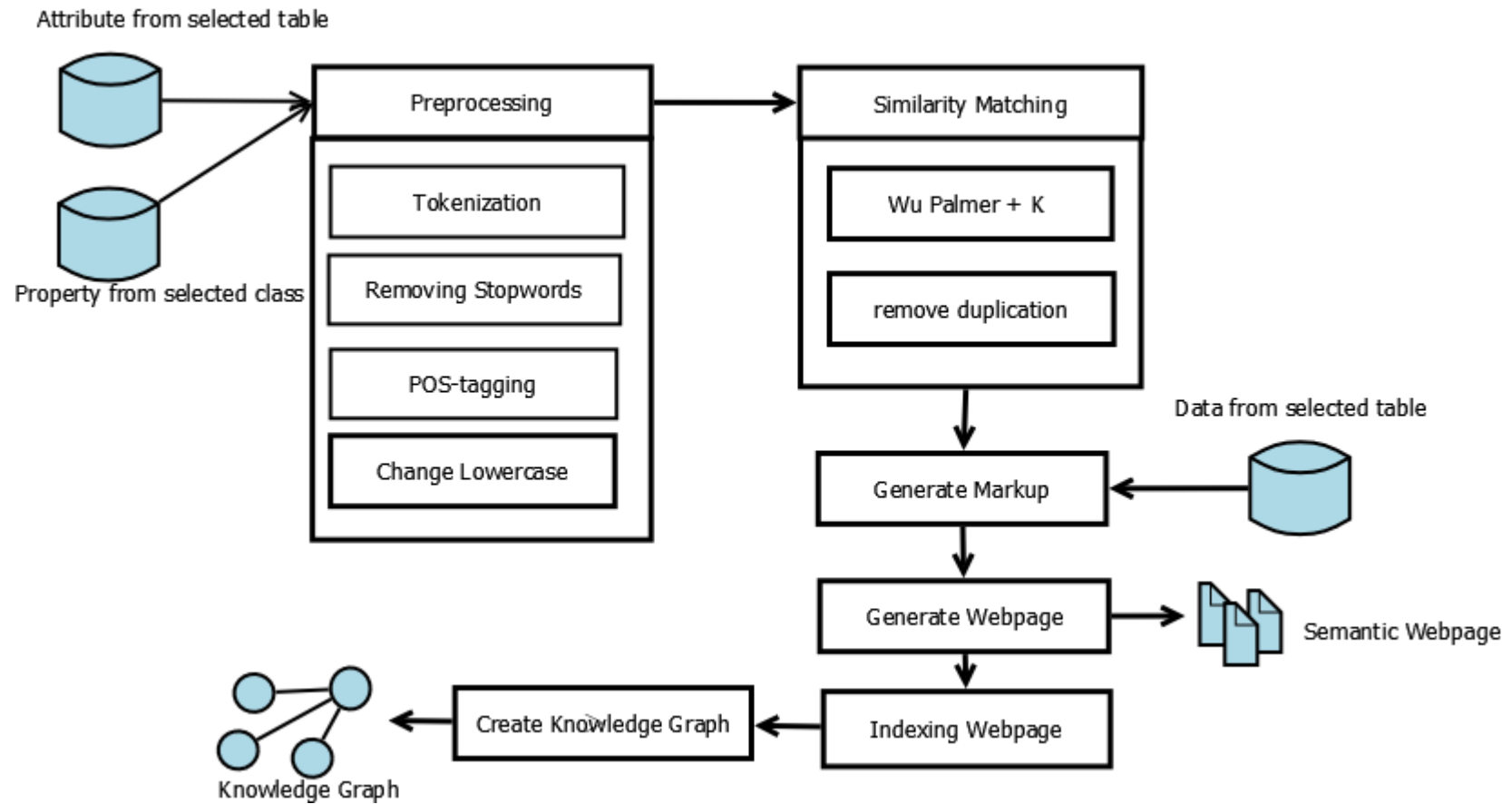
FoFA: Diet Information for Children with Autism with Semantic Technology in Android Based Application



Semantic Tagging

- “Semi Automatic Generating Semantic Markup Halaman Web Dari Data Terstruktur”*
- Propose the framework to develop webpages from structured data by using vocabulary schema.org
- Future work:
 - Make it as an automatic semantic tagging in broader domain
 - Building local knowledge graph (UNS)
 - Building local semantic search engine (UNS)

*Semi-Automatic Generating Semantic Markup Webpage from Structured Data with Semantic Matching





- “Automatic Semantic Tagging Teks Bahasa Indonesia Dengan N-Gram Language Model”*
- Algorithm to automatically create semantic tag (terms from ontologies) from dbpedia and schema.org of unstructured data
- Future Work:
 - Using the approach in the module integrated with CMS

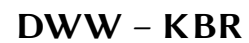
*) Automatic Semantic Annotation of Indonesian Language Phrase Using N-Gram Language Model.



Searching

- “Meningkatkan Hasil Question Answering System Dengan Metode Semantic Similarity Berbasis Hirarki Dalam Ontology”*
- Create framework in searching by using multiple ontologies
- Future Work:
 - Searching by using Linked Data on the Web

*) Improving the Result of Question Answering System with Semantic Similarity Method Based on Hierarchy in Ontology





- “Flexible Query Untuk Question Answering Dengan Template SPARQL Otomatis”*
- Composing algorithm for improving the result of NLP – SPARQL by creating template automatic (combination)
- Future Work
 - Utilizing ML to improve it

*) Automated SPARQL Template for Flexible Question Answering



- “Pemanfaatan SPARQL Inferencing Notation (SPIN) dalam Semantic Search. Studi Kasus : Tanaman Obat atau Ethnomedicine”
- SPIN is (almost) recommendation of Semantic Web, using new layer of Sem Web technology. SPARQL

*) SESS: Utilization of SPIN for Ethnomedicine Semantic Search



Populated Data

- “LS-MAPPING: Automatic Migration Dari Less Schema Relational Database Model (RDB) Ke Resource Description Framework (RDF)”*
- Providing algorithm to map not good structured data to RDF
- Future Work:
 - Accuracy is still low because syntax problem. It should be done first

*) Automatic Migration From Imperfect Relational Database To Resource Description Framework